

Nina Elmer

THEORETICAL PHYSICS · MACHINE LEARNING

Department of Applied Mathematics and Theoretical Physics, Cambridge, UK

Infosys - Cambridge AI Center, London, UK

✉ nme24@cam.ac.uk | 🏠 ninaelmer | 🌐 ninaelmer | in

Research Interest

- Physics and AI** Trustworthiness of AI models, Uncertainty estimation, BNNs, Ensembles, NTKs and GPs
- Particle Physics** Global analyses for SMEFT, PDF fitting, Amplitude surrogates

Experience

Research Associate

DAMTP, UNIVERSITY OF CAMBRIDGE

- Inaugural Physics-AI Fellow (independent postdoc), member of the Infosys-Cambridge AI Center

Cambridge, UK

Oct. 2025 - PRESENT

Postdoctoral Researcher

INSTITUTE FOR THEORETICAL PHYSICS, HEIDELBERG UNIVERSITY

- Continued research on the topic of amplification and uncertainty quantification for scattering amplitudes to finalise ongoing projects
- Advisor: Tilman Plehn

Heidelberg, Germany

Aug. 2025 - Sep. 2025

Education

PhD in Theoretical Particle Physics

HEIDELBERG UNIVERSITY

- Fellow of the IMPRS-PTFS at the Max-Planck-Institut für nukleare Physik
- Dissertation: "Bridging Theory and Data — Uncertainty-Aware Analyses for the LHC and Beyond"
- Supervisor: Tilman Plehn

Heidelberg, Germany

Nov. 2022 - July 2025

M.Sc in Physics

HEIDELBERG UNIVERSITY

- Thesis: "SMEFT Russian Roulette - Is including high kinematic distributions in SMEFT global analyses worth the effort?"
- Supervisor: Tilman Plehn

Heidelberg, Germany

Oct. 2020 - Sep. 2022

B.Sc in Physics

BIELEFELD UNIVERSITY

- Thesis: "Zusammenhang zwischen den Majorana-Phasen und der Massenordnung von Neutrinos"
- Supervisor: Dietrich Bödeker

Bielefeld, Germany

Oct. 2017 - Sep. 2020

Teaching

TUTORIALS

Heidelberg University

TEACHING ASSISTANT

- Classical Mechanics – Tutor (2022, 2023)
- Analytical Mechanics and Thermodynamics – Tutor (2023)
- Physics for Scientists – Tutor (2021, 2022)
- General Lab course – Tutor (2021)

Heidelberg, Germany

Nov. 2020 - Mar. 2024

Bielefeld University

TEACHING ASSISTANT

- General Lab course – Tutor (2019, 2020)

Bielefeld, Germany

Oct. 2019 - Sep. 2020

(CO-) SUPERVISION

Siqi Zheng

MPHIL IN DATA INTENSIVE SCIENCE (DIS)

- Project: Disentangling Uncertainties for LHC Amplitudes using Cooperative Bayesian and Variance Networks

Cambridge, UK

Nov. 2025 - July 2026

Giovanni De Crecenzo

PHD IN PHYSICS

- Project: Amplitude Surrogates for MADNIS

Heidelberg, Germany

Apr. 2025 - Present

Professional Service

SciPost Physics

REVIEWER/ REFEREE FOR SCIPOST PHYSICS

Online

Oct. 2025 - Present

Modern Numerics for Theoretical Physics

CO-ORGANIZER OF THE *Modern Numerics for Theoretical Physics* SEMINAR AT ITP HEIDELBERG

Heidelberg, Germany

Oct. 2024 - Aug. 2025

Fellowships

IMPRS-PFS Fellow

PHD FELLOWSHIP OF THE *International Max Planck Research School* IN HEIDELBERG

Heidelberg, Germany

Nov. 2022 - July 2025

Skills

Programming Python, C, C++, Fortran, Java, Basic SQL

Software and Tools PyTorch, TensorFlow, Mathematica, Git, scikit-learn, JAX

Languages German (native), English (fluent)

Talks

18 talks | 4 invited conference & workshop talks | 4 seminars

CONFERENCES & WORKSHOPS

General MadGraph Meeting

RELIABLE UNCERTAINTY ESTIMATION FOR AMPLITUDE SURROGATES

Heidelberg

Feb. 2026

Zurich Phenomenology Workshop

INVITED - LEARNING RELIABLE AMPLITUDE SURROGATES

Zurich

Jan. 2026

ACAT 2025 conference

LEARNING RELIABLE UNCERTAINTIES - THE RETURN OF THE ENSEMBLE

Hamburg

Sep. 2025

KISS BMBF Annual Meeting

INVITED — ACCURACY AND UNCERTAINTY CONTROL FOR ML-AMPLITUDES

Munich

Mar. 2025

MadGraph5_aMC@NLO Meeting

LEARNED UNCERTAINTIES FOR LHC-NETWORKS

CERN

Feb. 2025

ML4Jets Conference

THE GOOD, THE BAD, AND THE BAYESIAN

Paris

Nov. 2024

PHYSTAT – Statistics meet AI

POSTER — UNCERTAINTY-AWARE MACHINE LEARNING FOR THE LHC

London

Sep. 2024

IRN Terascale Meeting

THE IMPACT OF THEORY UNCERTAINTIES IN GLOBAL EDM ANALYSES

Frascati

Apr. 2024

Annual Meeting of the CRC TRR 257

INVITED — A GLOBAL INTERPRETATION OF EDMs

Karlsruhe

Mar. 2024

EDM Workshop

INVITED — A GLOBAL EDM ANALYSIS WITH THEORY UNCERTAINTIES

Trento

Mar. 2024

IRN Terascale Meeting

GLOBAL ANALYSIS: THE TRUE CONSTRAINING POWER OF DIFFERENT EDM MEASUREMENTS

Marseille

Oct. 2023

Pheno Conference

FROM HIGGS TO TOP: A GLOBAL SMEFT ANALYSIS

Pittsburgh

May 2023

ALPS Conference

PROFILING AND MARGINALIZATION IN A GLOBAL SMEFT ANALYSIS

Obergurgl

Mar. 2023

Higgs Conference

TO PROFILE OR TO MARGINALIZE: A SMEFT CASE STUDY

Pisa

Nov. 2022

DESY Theory Workshop

TO PROFILE OR TO MARGINALIZE: A SMEFT CASE STUDY

Hamburg

Sep. 2022

SEMINARS

COMETA Working Group Meeting

INVITED — USING AMPLITUDES FOR ACCURACY AND UNCERTAINTY ESTIMATION IN ML

Online

Jan. 2025

Nikhef Theory Seminar

INVITED — ACCURACY AND UNCERTAINTIES FOR ML-AMPLITUDES

Amsterdam

Dec. 2024

Particle Theory Seminar

INVITED — THE IMPACT OF LIKELIHOODS ON SMEFT GLOBAL ANALYSES

Würzburg

Oct. 2024

BNL Lunch Seminar

INVITED — FROM HIGGS TO TOP: A GLOBAL SMEFT ANALYSIS

Brookhaven

May 2023

Publications

Automatically updated list on Google Scholar

6 publications – 4 peer reviewed | 2 submitted

PEER-REVIEWED PUBLICATIONS

***Amplitude Uncertainties Everywhere All at Once**

WITH H. BAHL, T. PLEHN, R. WINTERHALDER

2025

SciPost Phys. 20, (2026) 083, arXiv:2509.00155 [hep-ph]

***Accurate Surrogate Amplitudes with Calibrated Uncertainties**

WITH H. BAHL, L. FAVARO, M. HAUßMANN, T. PLEHN, R. WINTERHALDER

2024

SciPost Phys. Core 8 (2025) 073, arXiv:2412.12069 [hep-ph]

***Staying on Top of SMEFT-Likelihood Analyses**

WITH M. MADIGAN, T. PLEHN, N. SCHMAL

2023

SciPost Phys. 18 (2025) 108, arXiv:2312.12502 [hep-ph]

***To profile or to marginalize — A SMEFT case study**

WITH I. BRIVIO, S. BRUGGESSER, E. GEOFFRAY, M. LUCHMANN, T. PLEHN

2022

SciPost Phys. 16 (2024) 035, arXiv:2208.08454 [hep-ph]

PREPRINT PUBLICATIONS

***Forecasting Generative Amplification**

WITH H. BAHL, S. DIEFENBACHER, T. PLEHN, J. SPINNER

2025

submitted to SciPost, arXiv:2509.08048 [hep-ph]

***A Global View of the EDM Landscape**

WITH S. DEGENKOLB, T. MODAK, M. MÜHLLLEITNER, T. PLEHN

2024

submitted to SciPost, arXiv:2403.02052 [hep-ph]